



## **CNS2105 DATA AND VOICE COMMUNICATIONS ASSIGNMENT TWO**

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1. Explain why it is necessary to modulate digital data onto analog carriers
2. (a) Shannon's channel capacity is given by  $C = B \log_2 \left(1 + \frac{S}{N}\right)$  with all the symbols having their usual meanings. Explain the trade-off between channel bandwidth and the signal power.  
(b) Consider a channel with a bandwidth of 1 MHz and an SNR of 127.

What is the theoretical upper limit to the data rate that the channel can carry?

- (c) Describe the following transmission impairments in communication systems and outline their causes:
  - i. Attenuation and attenuation distortion
  - ii. Delay distortion
3. Consider a communication link with a bandwidth of 20MHz and a signal to noise ratio of 30dB. Determine the theoretical channel capacity in bits per second.
4. A voice signal occupying a bandwidth of 0-4000Hz is to be transmitted over a digital link. Obtain the bandwidth of transmission under the following schemes:
  - a) PCM baseband transmission with 8-bit encoding of each sample
  - b) ASK transmission
  - c) M-ary FSK transmission with M=8